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The Aggregate Demand for Treasury Debt

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1 Big picture

- **Question.** Why do US Treasuries yield less than other seemingly safe assets? Is the gap mostly about default risk, or do Treasuries command a special premium because they are unusually *liquid* and *safe*?
- **Core idea.** The paper treats Treasuries as *convenience assets*, analogous to money. Investors value their liquidity and safety, so Treasury yields contain a *convenience yield*: they are lower than they would be in a frictionless asset-pricing model.
- **Setting.** The paper studies US annual data over a long historical sample, mainly 1919–2008 or 1926–2008 depending on the spread and controls. The empirical object is how changes in the market value of publicly held Treasury debt relative to GDP affect yield spreads between Treasuries and alternative assets.
- **Core challenge.** A Treasury–corporate spread reflects several things at once:
 - a convenience premium on Treasuries,
 - expected default losses on the non-Treasury asset, and
 - a default-risk premium.

So the paper must isolate the nondefault component of the spread and then separate *liquidity* from *safety*.

- **Main theoretical result.** In a representative-agent asset-pricing model augmented with convenience benefits from holding Treasuries and close substitutes, the Treasury convenience yield is decreasing in the supply of Treasuries (and related convenience assets) relative to GDP.
- **Main empirical result.** A fall in Treasury supply raises Treasury convenience yields sharply. In the log-spread regressions, a one-standard-deviation decrease in debt/GDP from its mean value 0.439 to 0.243 raises the Aaa–Treasury spread by about 44 basis points and the Baa–Treasury spread by about 77 basis points.
- **Main decomposition result.** Treasury supply moves spreads that isolate long-term safety (Baa minus Aaa corporate bonds), short-term safety (P2 minus P1 commercial paper), and liquidity (FDIC-insured CDs minus Treasury bills, and time/savings deposits minus Treasury bills). So Treasuries provide both *safety* and *liquidity* services.
- **Quantification.** Using a piecewise-linear specification, the paper estimates that the average convenience yield on long-term Treasuries over 1926–2008 is about **73 basis points**, of which at most **46 bps** are due to liquidity and at least **27 bps** are due to safety.
- **Economic takeaway.** Treasury debt is money-like. It delivers convenience services that lower yields, generate seigniorage-like gains for the government, and imply that observed Treasury rates are below the “true” riskless rate relevant for many asset-pricing and cost-of-capital calculations.

2 Data and measurement (key objects)

2.1 Observed data and sample windows

- **Frequency.** The regressions use annual observations.

- **Main sample horizon.** The longest spread regressions use data from 1919–2008 or 1926–2008. Shorter subsamples arise when specific control variables or asset series are unavailable.
- **Long-term corporate spreads.**
 - Moody’s Aaa long-maturity corporate bond yield minus long-maturity Treasury yield.
 - Moody’s Baa long-maturity corporate bond yield minus long-maturity Treasury yield.
- **Short-term spreads.**
 - high-grade commercial paper minus Treasury bills,
 - lower-grade (A2/P2) minus higher-grade (A1/P1) commercial paper, and
 - FDIC-insured CDs or time/savings deposits minus Treasury bills.
- **Treasury supply variable.** The key supply measure is the *market value* of publicly held US Treasury debt divided by GDP. The paper explicitly converts debt from face value to market value, because the theoretical object is the market value of outstanding convenience assets.
- **Maturity-specific Treasury supply.**
 - long-term Treasury supply: Treasuries with more than 10 years remaining maturity divided by GDP,
 - short-term Treasury supply: Treasuries with less than 1 year remaining maturity divided by GDP.

2.2 Key spread measures and what they are meant to capture

- **Aaa–Treasury spread.** A broad long-term spread with little default risk on the corporate side, used to show that Treasury supply matters even after strong default controls.
- **Baa–Treasury spread.** A broader long-term spread that should capture more of the full Treasury convenience yield, because Aaa bonds themselves may carry some convenience services.
- **Baa–Aaa spread.** Designed to isolate the *price of long-term safety*. Baa and Aaa bonds are similarly illiquid, but Baa bonds are riskier.
- **P2–P1 commercial paper spread.** Designed to isolate the *price of short-term safety*. The two commercial-paper classes are similarly illiquid, but P2 paper is riskier.
- **FDIC-insured CDs minus Treasury bills.** Designed to isolate the *price of liquidity*, because both are essentially safe over short horizons, but Treasuries are more liquid.
- **Time/savings deposits minus Treasury bills.** A longer historical liquidity proxy used for the pre-1984 period, though it is less clean than the FDIC-insured CD comparison.

2.3 Default-risk and risk-premium controls

- **EDF.** The Moody’s Analytics expected default frequency (EDF) for large firms, available beginning in 1969.
- **Stock return volatility.** Used as a longer-sample proxy for default risk when EDF is unavailable.

- **Yield-curve slope.** Measured as the spread between long-term Treasury yields and short-term Treasury bill yields. This is meant to capture business-cycle conditions and variation in risk premia.
- **Interpretation.** The paper’s claim is not merely that Treasury supply correlates with spreads because defaults rise in recessions. Instead, even after these default and risk-premium controls, the debt/GDP variable remains economically and statistically important.

2.4 Important empirical challenges

- **Measurement error in Treasury supply.** The convenience state variable in theory depends on Treasury supply plus close substitutes. Using debt/GDP alone introduces measurement error, which likely biases OLS coefficients *downward*.
- **Maturity composition is endogenous.** The government and private sector actively manage debt maturity, so long- or short-maturity Treasury supply may be endogenous. The paper addresses this with IV regressions.
- **Wartime intervention.** The 1940s and early 1950s are unusual because Federal Reserve wartime interventions affected Treasury pricing, especially at the short end. The authors are therefore cautious when interpreting level estimates of convenience yields from piecewise-linear specifications.

3 Models and empirical specifications (all equations + interpretation)

3.1 Convenience-yield representative-agent setup

The representative agent maximizes

$$\sum_{t=1}^{\infty} \beta^t u(C_t),$$

where consumption includes ordinary endowment plus convenience benefits:

$$C_t = c_t + n(v_t^A, GDP_t; y_t).$$

Here,

$$v_t^A = v_t^T + k^P v_t^P,$$

where:

- v_t^T is the market value of Treasuries,
- v_t^P is the market value of private-sector convenience assets, and
- k^P measures how much convenience these private substitutes provide relative to Treasuries.

Homogeneity of degree one lets the authors write

$$n(v_t^A, GDP_t; y_t) = GDP_t \cdot v\left(\frac{v_t^A}{GDP_t}; y_t\right).$$

Assumptions on the marginal convenience benefit:

$$v' \left(\frac{v_t^A}{GDP_t}; y_t \right) > 0, \quad v'' \left(\frac{v_t^A}{GDP_t}; y_t \right) < 0, \quad \lim_{v_t^A / GDP_t \rightarrow \infty} v' \left(\frac{v_t^A}{GDP_t}; y_t \right) = 0.$$

Interpretation. More Treasuries (and related safe/liquid substitutes) raise convenience, but with diminishing marginal convenience. That is exactly the object that should show up as a convenience yield in bond prices.

3.2 Pricing Treasuries and the basic convenience yield

Let the nominal pricing kernel be

$$M_{t+1} = \beta \frac{u'(C_{t+1})Q_t}{u'(C_t)Q_{t+1}},$$

where Q_t is the aggregate price level.

For a zero-coupon Treasury bond with nominal price P_t^T ,

$$P_t^T = \frac{E_t[M_{t+1}P_{t+1}^T]}{1 - v' \left(\frac{v_t^A}{GDP_t}; y_t \right)}.$$

Interpretation. The denominator is the source of the Treasury premium. A positive marginal convenience value raises the price of the Treasury bond and lowers its yield.

3.3 Short-term corporate–Treasury spread

For a one-period corporate bond, let default probability be λ_t and loss-given-default be $\tilde{L}_{t+1}$. Then the short spread is

$$S_{t,1} \equiv i_t^C - i_t^T = v' \left(\frac{v_t^A}{GDP_t}; y_t \right) + \lambda_t E_t[L_{t+1}] + \frac{\text{Cov}_t(M_{t+1}, \tilde{L}_{t+1})}{E_t[M_{t+1}]}.$$

Prediction 1. Holding default terms fixed, higher Treasury supply lowers the short corporate–Treasury spread.

Interpretation. The spread has three pieces:

1. the Treasury convenience yield,
2. expected default losses on the corporate bond, and
3. a default-risk premium.

This is the paper’s core identification logic: one must control for (2) and (3) to recover (1).

3.4 Long-term corporate–Treasury spread

For a t -period corporate–Treasury spread, the analogous expression is an average over the bond’s life:

$$S_{t,t} = \frac{1}{t} \sum_{j=t}^{t+t-1} E_t \left[v' \left(\frac{v_j^A}{GDP_j}; y_j \right) \right] + \frac{1}{t} \sum_{j=t}^{t+t-1} E_t[\lambda_j D_j] - \frac{1}{t} \sum_{j=t}^{t+t-1} \text{Cov}_t(m_{j+1}, R_{j+1}),$$

where the second term reflects expected default and the third term reflects the risk premium on corporate bonds relative to Treasuries.

Prediction 2. Higher Treasury supply lowers long-term corporate–Treasury spreads.

Interpretation. The long-term spread reflects the expected average convenience yield over the bond’s life. Since debt/GDP is persistent, long-term spreads are especially useful for recovering the supply-driven convenience component.

3.5 Separating liquidity from safety

The paper decomposes Treasury convenience into liquidity and safety components.

For short-term Treasuries:

$$v^{T,\text{short}}(\cdot) = v^{\text{liq}}\left(\frac{v_t^T + k^{\text{liq}}v_t^{P,\text{liq}}}{GDP_t}; y_t^{\text{liq}}\right) + v^{\text{short-safe}}\left(\frac{v_t^{T,\text{short}} + k^{\text{short-safe}}v_t^{P,\text{short-safe}}}{GDP_t}; y_t^{\text{short-safe}}\right).$$

For long-term Treasuries:

$$v^{T,\text{long}}(\cdot) = v^{\text{liq}}\left(\frac{v_t^T + k^{\text{liq}}v_t^{P,\text{liq}}}{GDP_t}; y_t^{\text{liq}}\right) + v^{\text{long-safe}}\left(\frac{v_t^{T,\text{long}} + k^{\text{long-safe}}v_t^{P,\text{long-safe}}}{GDP_t}; y_t^{\text{long-safe}}\right).$$

These decompositions generate three empirical predictions:

- **Prediction 3 (short-term safety).** If P2 and P1 commercial paper are similarly illiquid but differ in safety, then the P2–P1 spread should move with short-term Treasury supply.
- **Prediction 4 (long-term safety).** If Baa and Aaa corporate bonds are similarly illiquid but differ in safety, then the Baa–Aaa spread should move with long-term Treasury supply.
- **Prediction 5 (liquidity).** If FDIC-insured bank deposits and Treasury bills are similarly safe but differ in liquidity, then the deposit–bill spread should move with total Treasury supply.

Interpretation. This is the paper’s most important identification design. By comparing asset pairs that differ in only one attribute, the authors can say whether Treasury supply is changing the *price of safety* or the *price of liquidity*.

3.6 Baseline empirical specification

The baseline regressions take the form

$$\text{Spread}_t = \alpha + \beta \log\left(\frac{\text{debt}}{GDP}\right)_t + \Gamma X_t + \varepsilon_t,$$

where X_t includes default-risk and business-cycle controls such as EDF, stock-market volatility, and the slope of the Treasury yield curve.

- Regressions are estimated by OLS.
- Standard errors are adjusted for an AR(1) error structure.

- The authors prefer OLS with AR(1)-adjusted errors to GLS because measurement error in debt/GDP would be worse under a differenced GLS-type transformation.

Interpretation. The coefficient on $\log(\text{debt}/GDP)$ measures how Treasury supply affects the convenience component of the spread, net of default and risk-premium variation.

3.7 IV for maturity-specific Treasury supply

When the paper studies maturity-specific supply, it estimates IV regressions such as

$$\text{Spread}_t = \alpha + \beta \log\left(\frac{\text{long-term Treasuries}}{GDP}\right)_t + \Gamma X_t + \varepsilon_t,$$

or the analogous short-term supply regression.

The endogenous maturity-specific supply measure is instrumented with

$$\left(\frac{\text{debt}}{GDP}\right)_t, \quad \left(\frac{\text{debt}}{GDP}\right)_t^2, \quad \left(\frac{\text{debt}}{GDP}\right)_t^3.$$

Interpretation. The total amount of debt is used to generate exogenous variation in the maturity-specific component, because the maturity structure of debt itself may respond endogenously to market conditions.

3.8 Piecewise-linear specification for the level of the convenience yield

The log specification identifies the slope of the demand curve but not the absolute level of the convenience yield. To recover the level, the authors impose a function with an asymptote:

$$\text{Spread}_t = b_0 + b_1 \max\left\{b_2 - \frac{\text{debt}}{GDP_t}, 0\right\} + \Gamma X_t + \varepsilon_t.$$

- b_0 captures the default component of the spread (once controls are demeaned),
- the kinked term captures the convenience yield, and
- the convenience yield goes to zero once debt/GDP is sufficiently large.

Interpretation. This specification is what lets the authors say not just that Treasury supply matters, but by *how much* it lowers Treasury yields on average.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard spread comparisons miss

- A raw corporate–Treasury spread cannot be interpreted as convenience alone because it also contains expected default and a default-risk premium.
- Even a well-controlled long spread may still mix liquidity and safety.
- Treasury supply itself may be persistent and potentially correlated with macro conditions, so the paper needs controls and carefully chosen asset-pair comparisons.

4.2 Why debt/GDP is the right supply-side variable

- The theoretical convenience term depends on the stock of convenience assets relative to GDP, so the key regressor is the market value of publicly held Treasury debt divided by GDP.
- The paper focuses on *spreads* rather than on interest-rate levels. This removes many common shocks, such as expected inflation, that complicate inference in level regressions.
- The authors also argue that it is unlikely the government chooses the level of debt in response to the corporate bond spread itself, so reverse causality through this particular channel is less concerning.

4.3 How the paper separates convenience from default risk

- **Default controls.** EDF and stock-market volatility proxy for expected default risk. The yield-curve slope proxies for business-cycle conditions and time-varying risk premia.
- **Choice of spreads.** Short-term spreads, especially commercial-paper spreads, are less exposed to actual default events than long corporate spreads.
- **Asset-pair design.** Baa–Aaa and P2–P1 spreads hold liquidity roughly fixed and isolate safety. Deposit–bill spreads hold safety roughly fixed and isolate liquidity.

4.4 Why Aaa–Treasury is not enough

- Aaa bonds are very safe, but the paper argues they may themselves offer some convenience services.
- Therefore, the Aaa–Treasury spread understates the full Treasury convenience yield.
- This is why the Baa–Treasury spread is used as the main overall long-term measure of the convenience yield, while Baa–Aaa helps isolate the safety piece.

4.5 Why IV is used for long- and short-maturity Treasury supply

- The government’s maturity structure decision is potentially endogenous.
- Private-sector balance-sheet management may also move short- and long-duration safe-asset supplies endogenously.
- Using powers of total debt/GDP as instruments for maturity-specific Treasury supply helps recover the causal effect of supply on safety premia.

4.6 Caveats and limitations

- Piecewise-linear estimates of the *level* of the convenience yield are sensitive to the wartime period because the Federal Reserve intervened heavily in Treasury markets in the 1940s.
- Short-term convenience-yield *levels* are harder to pin down because wartime interventions were especially strong in bills.

- The reduced-form coefficients measure the effect of Treasury supply *net of any response in private-sector substitute assets*; they do not separately identify the full structural demand system for all convenience assets.

4.7 Relation to the literature

- Relative to the corporate bond pricing literature, the paper adds direct evidence that a substantial nondefault component of corporate–Treasury spreads is driven by Treasury supply.
- Relative to earlier Treasury supply papers, it studies a much longer sample, uses several ways to rule out default explanations, and explicitly decomposes the convenience yield into safety and liquidity.
- Relative to standard representative-agent risk-premium stories, the paper emphasizes that quantities of safe/liquid assets matter for prices because Treasuries provide convenience services.

5 Tables and figures

5.1 Figure 1: Corporate bond spreads and government debt

Read:

- The figure plots the Aaa–Treasury spread against debt/GDP from 1919 to 2008.
- The relationship is strongly negative: when Treasury supply is low, the spread is high.
- This is the paper’s visual money-demand style fact: scarce Treasuries command a bigger convenience premium.
- The figure is not by itself causal evidence, but it is the central reduced-form motivation for the rest of the paper.

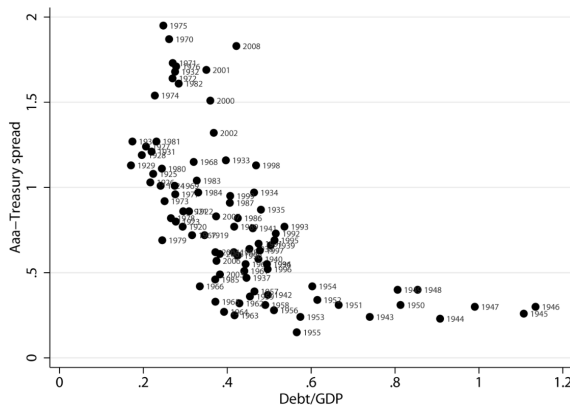


FIG. 1.—Corporate bond spread and government debt. The figure plots the Aaa–Treasury corporate bond spread (y axis) against the debt-to-GDP ratio (x axis) on the basis of annual observations from 1919 to 2008. The corporate bond spread is the difference between the percentage yield on Moody’s Aaa long-maturity bond index and the percentage yield on long-maturity Treasury bonds.

	A. AAA-TREASURY			B. BAA-TREASURY		C. CP-BILLS		D. CFP2-BILLS	
	1919–2008	1969–2008	1926–2008	1969–2008	1926–2008	1920–2008	1969–2008	1926–2008	1974–2007
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log(debt/GDP)	-.746	-.870	-.800	-1.662	-1.309	-.730	-.961	-.354	-1.958
	[–4.36]	[–2.78]	[–5.12]	[–3.96]	[–7.55]	[–4.42]	[–2.08]	[–3.56]	[–5.97]
EDF		1.040		1.433			.126		.040
		[3.51]		[3.21]			[.24]		[.08]
Volatility			1.313		6.344			1.902	
			[1.93]		[6.81]			[2.28]	
Slope		.058	.081	.221	.312		–.102	–.083	–.10
		[1.26]	[1.89]	[2.88]	[4.69]		[–1.09]	[–1.39]	[–1.10]
Intercept	.109	.118	.077	.377	.737	.095	–.200	.227	–.849
	[.62]	[.35]	[.49]	[.84]	[4.33]	[.57]	[–.40]	[1.48]	[–1.66]
R ²	.449	.599	.573	.577	.690	.226	.184	.258	.287
ρ	.694	.660	.532	.171	.013	.175	–.022	.016	.074
Observations	90	40	83	40	83	89	40	83	34

NOTE.—The dependent variables are long- and short-term yield spreads between corporate and Treasury bonds, both measured in percentage units. Independent variables are the log of the ratio of the market value of Treasury debt outstanding to US GDP and controls for the default risk and default risk premium on corporate bonds. EDF is the expected default frequency for corporate bonds. Volatility is the annualized standard deviation of weekly log stock returns on the S&P 500 index. Slope is the slope of the Treasury yield curve measured as the spread between the 10-year Treasury yield and the 3-month Treasury yield. Appendix B provides the precise definitions of all variables. EDF, volatility, and slope controls are demeaned. Regressions are estimated by OLS. The standard errors are adjusted assuming that errors are AR(1). We use the Box-Jenkins methodology for identifying the error structure. ρ denotes the first-order autocorrelation of the error terms. *t*-statistics are in brackets.

5.2 Table 1: Baseline impact of Treasury supply on spreads

Read:

- For the Aaa–Treasury spread, the baseline log specification gives a coefficient of about -0.746 on $\log(\text{debt}/GDP)$. This implies that a one-standard-deviation fall in debt/GDP from 0.439 to 0.243 raises the spread by about **44 basis points**.
- The result survives the inclusion of EDF, stock-return volatility, and the slope of the yield curve. So it is not just a story about default risk or business-cycle risk premia.
- For the Baa–Treasury spread, the coefficient is larger, around -1.309 in the longer sample with volatility controls. The implied effect of the same one-standard-deviation fall in debt/GDP is about **77 basis points**.
- The larger Baa–Treasury response is important: it suggests Aaa bonds themselves provide some convenience, so the Baa–Treasury spread is closer to the full convenience-yield object.
- Short spreads behave similarly. Treasury supply also lowers CP–bill and P2/P1-related spreads, showing that the effect is not unique to long corporate bonds.

5.3 Table 2: Evidence for priced safety and priced liquidity

Read:

- **Long-term safety.** The Baa–Aaa spread falls when Treasury supply rises. In the IV regression using long-term Treasury supply, the coefficient is about -0.304 , implying that a one-standard-deviation fall in long-term Treasury supply raises the *price of long-term safety* by about **40 basis points**.
- **Short-term safety.** The P2–P1 commercial-paper spread also falls when Treasury supply rises. In the IV regression using short-term Treasury supply, the coefficient is about -1.462 , implying that a one-standard-deviation fall in short-term Treasury supply raises the *price of short-term safety* by about **26 basis points**.
- **Liquidity.** The spread between FDIC-insured six-month CDs and six-month Treasury bills is strongly decreasing in Treasury supply, with a coefficient around -1.884 . This is direct evidence that the *price of liquidity* rises when Treasuries are scarce.
- The older time/savings-deposit evidence also points in the same direction, though it is less clean than the FDIC-insured CD comparison.
- This table is the key identification table: it shows that the Treasury convenience yield is not a single black box but the sum of a safety premium and a liquidity premium.

5.4 Table 3: Quantifying the level of the convenience yield

Read:

- The piecewise-linear specification imposes that the convenience yield goes to zero when Treasury supply is sufficiently large.

TABLE 2
IMPACT OF TREASURY SUPPLY ON THE PRICE OF SAFETY AND THE PRICE OF LIQUIDITY

	A. PRICE OF SAFETY				B. PRICE OF LIQUIDITY	
	$S^{\text{Baa-Aaa}}$		$S^{\text{P2-P1}}$		$S^{\text{FDIC-insured CDs-Bills}}$	$S^{\text{Time/Savings/Accounts-Bills}}$
	1926–2008 (1)	1926–2008 (2)	1974–2007 (3)	1974–2007 (4)	1984–2008 (5)	1935–65 (6)
Log(debt/GDP)	–.510 [–3.45]		–.888 [–4.34]		–1.884 [–1.71]	–.690 [–2.57]
Log(debt > 10-year maturity/GDP)*		–.304 [–2.44]				
Log(debt ≤ 1-year maturity/GDP)*				–1.462 [–2.84]		
Volatility	5.031 [6.47]	6.232 [6.65]				
EDF			.074 [.35]	.019 [.07]		
Slope	.232 [4.20]	.211 [3.29]	.014 [.42]	.055 [1.13]	.101 [.88]	1.181 [10.04]
Constant	.660 [4.54]	.264 [.74]	–.508 [–2.39]	–2.679 [–2.47]	–1.486 [–1.52]	–.170 [–.95]
Observations	83	83	34	34	25	31
R^2	.600		.497		.233	.898
Estimation method	OLS	IV	OLS	IV	OLS	OLS
Error term	AR(1)	AR(1)	AR(1)	AR(1)	AR(1)	AR(1)

NOTE.—The dependent variables are the spread between Baa-rated corporate bonds and Aaa-rated bonds (cols. 1 and 2), the spread between A2/P2-rated and A1/P1-rated commercial paper (cols. 3 and 4), the spread between the interest rate on FDIC-insured 6-month CDs and 6-month Treasury bills (col. 5), and the spread between the average interest rate paid by banks on time and savings deposits and 6-month Treasury bills (col. 6). Independent variables are the log of the ratio of the market value of Treasury debt outstanding to US GDP, the log of the ratio of the market value of Treasury debt with remaining maturity greater than 10 years to US GDP, the log of the ratio of the market value of Treasury debt with remaining maturity less than 1 year to US GDP, the annualized standard deviation of weekly log stock returns on the S&P 500 index (volatility), the expected default frequency for corporate bonds (EDF), and the slope of the Treasury yield curve measured as the spread between the 10-year Treasury yield and the 3-month Treasury yield in cols. 1–4 and as the spread between the 10-year Treasury yield and the 6-month Treasury yield in cols. 5 and 6. Appendix B provides the precise definitions of all variables. EDF, volatility, and slope controls are demeaned. Regressions are estimated by OLS or instrumental variables (two-stage least squares), where we use instruments debt/GDP, (debt/GDP)², and (debt/GDP)³, as indicated. The standard errors are adjusted assuming that errors are AR(1). We use the Box-Jenkins methodology for identifying the error structure. *t*-statistics are in brackets.

* Instrumented by (debt/GDP), (debt/GDP)², and (debt/GDP)³.

TABLE 3
IMPACT OF TREASURY SUPPLY ON BOND SPREADS: PIECEWISE-LINEAR SPECIFICATION

	A. AAA-TREASURY		B. BAA-TREASURY	
	1919–2008 (1)	1926–2008 (2)	1919–2008 (3)	1926–2008 (4)
b_0	.319 [1.80]	.347 [2.55]	1.019 [1.94]	1.200 [7.68]
b_1	2.590 [4.06]	3.075 [5.17]	4.309 [2.64]	4.952 [6.76]
b_2	.584 [7.02]	.548 [9.74]	.626 [4.22]	.546 [12.92]
Volatility		1.206 [1.93]		6.215 [6.98]
Slope		.096 [2.43]		.334 [5.07]
R^2	.479	.618	.290	.704
Observations	90	83	90	83

NOTE.—This table estimates a piecewise-linear specification for the relation between Treasury supply and bond spreads. The function estimated is $b_0 + b_1 \times \max[b_2 - \text{debt/GDP}, 0]$. The dependent variables are long-term yield spreads between corporate and Treasury bonds, measured in percentage units. Independent variables are the ratio of the market value of Treasury debt outstanding to US GDP and controls for the default risk and default risk premium on corporate bonds. Volatility is the annualized standard deviation of weekly log stock returns on the S&P 500 index. Slope is the slope of the Treasury yield curve measured as the spread between the 10-year Treasury yield and the 3-month Treasury yield. Appendix B provides the precise definitions of all variables. Volatility and slope are demeaned. Regressions are estimated by nonlinear least squares. The standard errors are adjusted assuming that errors are AR(1). We use the Box-Jenkins methodology for identifying the error structure. *t*-statistics are in brackets.

- Using the Baa–Treasury spread, the paper estimates an *average long-term Treasury convenience yield* of about **73 basis points** over 1926–2008.
- Using the Aaa–Treasury spread as a benchmark for liquidity gives an upper bound of about **46 basis points** for the liquidity component.
- The residual then implies a lower bound of about **27 basis points** for the safety component.
- The table therefore turns the regression slopes into an economically interpretable level estimate.

5.5 Conclusion calculations and asset-pricing implications

Read:

- Multiplying the convenience yield by the historical average debt/GDP ratio implies that the government earns seigniorage-like gains of roughly **0.23–0.25 percent of GDP per year**.
- This is about the same order of magnitude as traditional monetary seigniorage from the public’s willingness to hold fiat money at low interest.
- The paper argues that Treasury yields are therefore not the correct “true” riskless rate for many asset-pricing applications.

6 Short summary

- **Method.** The paper adds convenience benefits from holding Treasuries to a standard asset-pricing framework and maps the model into short- and long-maturity yield-spread regressions.
- **Identification.** It uses Treasury supply variation together with default controls and carefully chosen asset pairs to separate the convenience yield from default risk and then decompose convenience into *safety* and *liquidity*.
- **Main result.** Treasury scarcity lowers Treasury yields substantially. A one-standard-deviation fall in debt/GDP raises long corporate–Treasury spreads by roughly 44 to 77 basis points, depending on the corporate bond used.
- **Quantification.** The average convenience yield on long-term Treasuries is about 73 basis points, with at most 46 basis points due to liquidity and at least 27 due to safety.
- **Implication.** Treasuries are special because they are money-like safe and liquid assets. Their yields are therefore below the true riskless rate, and the government captures meaningful seigniorage from issuing them.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Treasury debt is in special demand because it provides convenience services beyond ordinary pecuniary payoffs.
- The amount of Treasury debt outstanding has a large causal effect on yield spreads relative to other assets.
- Both *liquidity* and *safety* are important parts of this convenience premium.

7.2 Economic intuition

- When Treasuries are scarce, investors bid them up because they are especially easy to trade and especially safe to hold or post as collateral.
- Since these convenience services are valuable, the Treasury yield is pushed below the yield on less convenient assets.
- As Treasury supply increases, these services become less scarce, so the convenience yield shrinks and spreads against other assets narrow.

7.3 Takeaway

This paper is important because it changes how to think about Treasury rates. Treasuries are not just default-free discount bonds; they are also special safe and liquid instruments that investors value directly. That insight helps explain large nondefault components in corporate–Treasury spreads, gives the government a seigniorage-like benefit from debt issuance, and implies that many asset-pricing exercises should adjust Treasury yields upward to recover the true riskless rate.